Functional Requirements | Structured Edition

RF01–RF36 • Workflow-aligned • Functional requirements and verification criteria in bullet form. Content follows the consolidated source edition.

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Data Harvesting

RF01 | Search

Functional requirements

- The application shall allow users to specify search criteria, including geographical area, time period, and biological group.

Verification criteria

- Enter valid filters and verify that they are correctly captured by the GUI and passed to the search workflow.

RF02 | GBIF Data Harvesting

Functional requirements

- The application shall retrieve biodiversity occurrence data from the Global Biodiversity Information Facility (GBIF) through its supported APIs, using a hybrid harvesting approach that accommodates both direct record retrieval and asynchronous dataset download.
- The application shall automatically select the appropriate retrieval mechanism, including the GBIF Occurrence Search API or Occurrence Download API, according to documented deterministic rules considering query compatibility, expected result volume, API limitations, authentication requirements, and requested output characteristics.
- The selected retrieval mechanism and its execution status shall be visible to the user.

- The application shall verify the technical completeness and integrity of retrieval operations, including successful response handling, pagination or download completion, and consistency between expected and retrieved records where determinable.
- Retrieved data shall be preserved without scientific corrections or semantic harmonisation during harvesting.
- The application shall retain partially retrieved data when acquisition is interrupted or incomplete, explicitly marking the corresponding harvesting operation and its outputs as incomplete.
- Incomplete acquisitions shall not be treated as successfully completed datasets or passed to Data Validation as complete acquisitions.
- The application shall provide sufficient information to identify the incomplete operation and support subsequent recovery or re-execution.
- The application shall record structured harvesting provenance, including GBIF source identifiers, original record and dataset identifiers where available, executed query parameters, retrieval timestamps, selected API mechanism, download keys and DOIs where available, acquisition status, and relevant expected and retrieved record counts.
- The recorded information shall support traceability of harvested data to their original GBIF sources and retrieval operations.

Verification criteria

- Execute predefined GBIF harvesting scenarios involving queries compatible with direct record retrieval, queries requiring asynchronous downloads because of result volume or output requirements, authenticated download operations, paginated responses, empty results, and interrupted or incomplete acquisitions.
- Verify that the application automatically selects the appropriate GBIF retrieval mechanism according to documented deterministic rules and correctly displays the selected mechanism and execution status to the user.
- Verify that supported query parameters are correctly transmitted to the selected GBIF API and that authentication requirements are handled when applicable.
- Verify that the application retrieves the expected occurrence records or download outputs, correctly handles pagination and asynchronous download completion, and checks technical retrieval completeness and integrity.
- Verify that the harvested records retain their original values without scientific correction or semantic harmonisation.
- Simulate interrupted pagination, incomplete downloads, and other partial retrieval conditions.
- Verify that successfully retrieved data are retained, the affected acquisition is explicitly classified as incomplete, and incomplete outputs are not passed to Data Validation as completed acquisitions.
- Verify that the application preserves sufficient information to identify and subsequently recover or re-execute incomplete operations.
- Verify that harvesting provenance records include available GBIF source, dataset, and record identifiers, executed query parameters, retrieval timestamps, selected retrieval mechanism, download keys and DOIs where available, acquisition status, and relevant record counts.
- Verify that harvested records can be traced to their original GBIF sources and associated retrieval operations.

RF03 | OBIS Data Harvesting

Functional requirements

- The application shall retrieve marine biodiversity data and associated metadata from the Ocean Biodiversity Information System (OBIS) through its supported data access services, using a hybrid harvesting approach that accommodates direct API-based record retrieval and dataset-level acquisition.
- The application shall automatically select an appropriate retrieval mechanism according to documented deterministic rules considering query compatibility, expected data volume, available data access methods,

requested content, and structural characteristics.

- The selected mechanism and the status of the harvesting operation shall be visible to the user.
- The application shall preserve the structure of harvested data as provided by the selected OBIS service.
- Where available, separate Event, Occurrence, and ExtendedMeasurementOrFact (eMoF) tables, their identifiers, and documented relationships shall be retained.
- Where data are retrieved in a flattened representation, the application shall preserve that representation without reconstructing unsupported relationships.
- The application shall explicitly manage the inclusion or exclusion of absence records and records excluded by OBIS quality- control procedures, where these records are available through supported access mechanisms.
- The application shall inform users about applicable default exclusions, supported inclusion options, and any limitations of the selected retrieval mechanism.
- User-defined inclusion criteria shall be respected and recorded.
- The application shall preserve the retrieved OBIS data without introducing scientific corrections or semantic harmonisation during harvesting.
- It shall distinguish OBIS-processed or enriched information from original publisher-provided or verbatim information where such distinctions are available in the retrieved data or associated metadata, without inferring unsupported original values.
- The application shall verify the technical completeness and integrity of retrieval operations, including response handling, pagination or resource acquisition completion, and consistency between expected and retrieved outputs where determinable.
- Partially retrieved data shall be retained and explicitly marked as incomplete, without being passed to Data Validation as completed acquisitions.
- The application shall record structured harvesting provenance, including OBIS source and dataset identifiers, original record identifiers and persistent identifiers where available, executed query parameters, retrieval timestamps, selected access mechanism, inclusion and exclusion criteria, retrieval status, and relevant record counts.
- The recorded information shall support traceability to the original OBIS resources and harvesting operations.

Verification criteria

- Execute predefined OBIS harvesting scenarios involving direct occurrence retrieval, dataset-level acquisition, different result volumes, structured and flattened data representations, empty results, and interrupted or incomplete retrieval operations.
- Verify that the application automatically selects an appropriate OBIS retrieval mechanism according to documented deterministic rules and displays the selected mechanism and execution status.
- Verify that supported query parameters are correctly transmitted and that unsupported parameters or access limitations are explicitly reported.
- Provide test resources containing separate Event, Occurrence, and ExtendedMeasurementOrFact tables with documented identifiers and relationships.
- Verify that the application preserves the original structures and relationships.
- Provide flattened records with insufficient relational information and verify that the application does not invent or reconstruct unsupported relationships.
- Execute harvesting scenarios involving presence records, absence records, and records excluded by OBIS quality- control procedures, using access mechanisms that support the relevant options.
- Verify that applicable default exclusions and available inclusion options are communicated to the user, that user-defined inclusion criteria are respected, and that the applied criteria are recorded in harvesting provenance.
- Verify that unavailable inclusion options are explicitly reported rather than silently ignored.

- Provide records containing OBIS-processed or enriched values alongside available publisher-provided or verbatim information.
- Verify that the application preserves the retrieved values and correctly distinguishes their origins where documented, without inventing original values or modifying scientific content.
- Verify that the application detects technical retrieval failures and incomplete acquisitions, retains successfully retrieved data, and explicitly marks incomplete outputs.
- Verify that incomplete acquisitions are not passed to Data Validation as completed datasets and that sufficient information is retained to support recovery or re- execution.
- Verify that harvesting provenance includes available OBIS source, dataset, record, and persistent identifiers, executed queries, retrieval timestamps, selected access mechanisms, inclusion and exclusion criteria, retrieval status, and relevant record counts.
- Verify that harvested outputs remain traceable to their original OBIS resources and retrieval operations.

RF04 | Figshare Data Harvesting

Functional requirements

- The application shall support the discovery, selection, and retrieval of research datasets, associated files, and metadata from Figshare through its supported data access services, using a hybrid harvesting approach that accommodates metadata retrieval and file-level acquisition.
- The application shall execute structured searches using supported query parameters and retrieve available descriptive and technical metadata, including dataset titles, descriptions, authorship, keywords, resource identifiers, persistent identifiers, publication information, file descriptions, version information, and access and licensing conditions where available.
- The application shall assess the potential relevance of discovered resources to the user's research criteria through deterministic filtering and AI-assisted recommendations.
- AI-assisted assessments shall provide contextual explanations based on available metadata and shall distinguish supported evidence from uncertainty.
- AI-generated relevance assessments shall remain advisory and shall not automatically determine which datasets or files are acquired or excluded.
- The application shall present discovered resources and their associated files to the user and require explicit user selection and approval before initiating data file downloads.
- The application shall preserve the distinction between resource discovery, metadata harvesting, and acquisition of selected files.
- The application shall retrieve and preserve user-selected files in their original formats, independently of whether those formats are currently supported for parsing, validation, or subsequent processing.
- The application shall identify available file characteristics without altering the original content, inferring unsupported structures, or treating technical acquisition as evidence of scientific validity or downstream compatibility.
- The application shall identify and record the specific resource versions and file identifiers associated with each acquisition, together with available access restrictions, licensing information, and persistent identifiers.
- Where a selected resource or file is inaccessible, unavailable, or subject to unmet access requirements, the application shall report the limitation and preserve any successfully retrieved metadata.
- The application shall verify the technical completeness and integrity of retrieval operations, including response handling, pagination, file availability, and download completion.
- Partially retrieved files or resources shall be retained where technically possible and explicitly marked as incomplete, without being passed to Data Validation as completed acquisitions.
- The application shall record structured harvesting provenance, including Figshare resource and file identifiers, persistent identifiers where available, selected versions, executed queries, retrieval timestamps, applied

relevance criteria, AI-assisted recommendations and user decisions, access mechanisms, retrieval status, and relevant resource or file counts.

- The recorded information shall support traceability to the original Figshare resources and harvesting operations.

Verification criteria

- Execute predefined Figshare harvesting scenarios involving structured resource searches, multiple matching results, heterogeneous file formats, different resource versions, publicly accessible and restricted resources, empty results, and interrupted or incomplete acquisitions.
- Verify that the application correctly executes supported search parameters and retrieves available descriptive and technical metadata, including resource identifiers, persistent identifiers, authorship, publication information, file descriptions, version information, and licensing conditions.
- Verify that unsupported search criteria are explicitly reported rather than silently ignored.
- Provide resources with varying degrees of relevance to predefined research criteria.
- Verify that deterministic filtering and AI-assisted relevance assessments correctly distinguish documented evidence from uncertainty, provide understandable recommendations, and do not automatically exclude or select resources based solely on AI-generated assessments.
- Verify that the application presents discovered datasets and associated files for explicit user selection and approval before initiating file downloads.
- Confirm that unselected files are not downloaded and that the application records the user's selection decisions.
- Provide selected files in both supported and unsupported formats.
- Verify that all accessible selected files can be acquired and preserved in their original formats without content modification, automatic structural interpretation, or unsupported assumptions about downstream compatibility.
- Provide multiple versions of the same resource and verify that the application correctly identifies and records the acquired version and associated file identifiers.
- Simulate inaccessible files and unmet access requirements and verify that the application reports the limitations while preserving successfully retrieved metadata.
- Simulate interrupted searches, paginated responses, failed file downloads, and partial acquisitions.
- Verify that the application correctly identifies incomplete operations, retains available partial outputs where technically possible, and prevents incomplete acquisitions from being treated as completed inputs to Data Validation.
- Verify that harvesting provenance records include available Figshare resource and file identifiers, persistent identifiers, resource versions, executed queries, retrieval timestamps, applied relevance criteria, AI-assisted recommendations, user selection decisions, access mechanisms, retrieval status, and relevant counts.
- Verify that each acquired file can be traced to its corresponding Figshare resource, version, and harvesting operation.

RF05 | eLTER Data Harvesting

Functional requirements

- The application shall support the discovery, selection, and retrieval of ecological research sites, datasets, digital assets, and associated metadata from the European Long-Term Ecosystem Research infrastructure (eLTER), using a hybrid harvesting approach that accommodates site-level information retrieval, dataset and asset discovery, and acquisition of accessible data resources.
- The application shall integrate DEIMS-SDR as a source of research site information and the eLTER Digital Asset Register (DAR) as a source of dataset and digital asset metadata, using their available and supported data access interfaces.

- The application shall distinguish site identification and description from dataset discovery and the retrieval of observational data.
- The application shall execute supported structured searches and retrieve available information concerning site identifiers, geographic location, ecosystem characteristics, observed variables, temporal coverage, dataset descriptions, methodologies, resource identifiers, access conditions, and licensing information, where available and applicable.
- The application shall preserve documented relationships between eLTER sites and datasets or digital assets, including their original identifiers and references.
- Where associations are not explicitly documented, the application shall not automatically establish them based solely on geographic proximity, temporal overlap, or other indirect similarities.
- The application shall support deterministic relevance assessment and AI-assisted recommendations for potentially relevant datasets and possible site–dataset associations.
- AI- assisted recommendations shall provide supporting evidence and indicate uncertainty.
- Such recommendations shall remain advisory and shall not be incorporated as verified relationships without explicit user approval and sufficient supporting evidence.
- The application shall present discovered datasets and digital assets for explicit user selection and approval before initiating file acquisition.
- It shall retrieve accessible user-selected resources through supported access mechanisms and preserve acquired files in their original formats without introducing scientific corrections, semantic harmonisation, or unsupported structural transformations.
- Where automated file retrieval is unavailable, restricted, or unsupported, the application shall retain the available metadata, resource identifiers, access information, and references to the original resources.
- It shall explicitly distinguish resources that have been discovered or described from resources whose underlying data files have been successfully acquired.
- The application shall verify the technical completeness and integrity of harvesting operations, including response handling, resource availability, and retrieval completion.
- Partially acquired data shall be retained where technically possible and explicitly marked as incomplete, without being passed to Data Validation as completed acquisitions.
- The application shall record structured harvesting provenance, including source services, site and resource identifiers, documented site– dataset associations, executed queries, retrieval timestamps, access mechanisms, relevance assessments, AI-assisted recommendations, user decisions, acquisition status, and available persistent identifiers.
- The recorded information shall support traceability to the original eLTER services, sites, datasets, and harvesting operations.

Verification criteria

- Execute predefined eLTER harvesting scenarios involving site discovery through DEIMS- SDR, dataset and digital asset discovery through the eLTER Digital Asset Register, retrieval of available metadata, and acquisition of accessible resources.
- Include scenarios involving complete, incomplete, restricted, and unavailable data resources.
- Verify that the application correctly distinguishes site- level information from dataset and digital asset metadata and observational data.
- Verify that supported query parameters are correctly transmitted to the relevant services and that unsupported search criteria or access limitations are explicitly reported.
- Provide test resources containing documented site– dataset associations and resources without documented associations.
- Verify that the application preserves existing relationships and original identifiers without creating unsupported links.

- Verify that geographic proximity, temporal overlap, or similar characteristics do not independently result in automatic establishment of verified associations.
- Provide resources with different degrees of relevance to predefined ecological research criteria.
- Verify that deterministic assessments and AI-assisted recommendations are accompanied by supporting evidence and uncertainty information.
- Verify that proposed site–dataset associations remain unverified until explicitly approved by the user and supported by sufficient evidence.
- Verify that discovered datasets and digital assets are presented for explicit user selection before file acquisition.
- Confirm that only approved files are downloaded and that acquired files retain their original formats and content.
- Simulate resources that are discoverable through the eLTER services but whose data files cannot be retrieved automatically.
- Verify that the application preserves available metadata, identifiers, access conditions, and source references, and clearly distinguishes discovered resources from successfully acquired files.
- Simulate interrupted retrieval operations, inaccessible resources, and incomplete acquisitions.
- Verify that the application correctly reports acquisition status, retains available partial outputs where technically possible, and prevents incomplete acquisitions from being treated as completed inputs to Data Validation.
- Verify that harvesting provenance includes source services, site and resource identifiers, documented associations, executed queries, retrieval timestamps, selected access mechanisms, relevance assessments, AI recommendations, user decisions, acquisition status, and available persistent identifiers.
- Verify that each harvested resource can be traced to its original eLTER source and associated harvesting operation.

RF06 | Harvested Data Provenance

Functional requirements

- The application shall implement a common provenance management framework to preserve, organise, and maintain traceability of data, metadata, and digital resources harvested from all supported sources, including GBIF, OBIS, Figshare, and eLTER.
- The application shall record provenance at source, harvesting operation, and resource levels, extending traceability to individual records and attributes where necessary and technically feasible.
- Provenance information shall include available source identifiers, original record and resource identifiers, persistent identifiers, source versions, retrieval parameters, timestamps, access mechanisms, acquisition statuses, and relevant technical information.
- The application shall assign a unique internal identifier to each harvesting operation and record every acquisition as a distinct event, including repeated acquisitions of the same source dataset or resource.
- The application shall preserve the relationship between repeated acquisitions and their common source without overwriting previous acquisition histories.
- The application shall preserve original source identifiers without modification and maintain explicit mappings between source identifiers and internally assigned identifiers.
- Where original identifiers are unavailable, the application shall generate internal identifiers while clearly distinguishing them from identifiers provided by the source.
- The application shall maintain explicit relationships between harvested resources, their source datasets, associated files, individual records where identifiable, and the harvesting operations through which they were acquired.

- These relationships shall support traceability without introducing undocumented associations between independently acquired resources.
- The application shall record available technical integrity information, including file sizes, checksums, retrieval counts, and acquisition completion indicators where applicable.
- It shall preserve provenance information for completed, incomplete, and failed harvesting operations, including relevant errors and recovery information.
- The application shall distinguish deterministic processing, AI- generated recommendations, and user decisions in provenance records.
- Where AI-assisted assessments or user approvals influence resource selection or subsequent processing, the application shall record their relevant context, outcomes, and relationships to the affected resources or operations.
- The application shall provide a common provenance and lineage model that supports traceability across Data Validation, Data Harmonisation, Data Integration, Data and Metadata Export, and subsequent processing stages.
- Each processing module shall be responsible for recording its own operations, inputs, outputs, modifications, decisions, and applicable validation results using the shared model.
- The application shall preserve links between original harvested data and derived outputs, including record-level and attribute-level relationships where necessary and technically feasible.
- Derived values shall remain traceable to their original inputs and documented transformations, without modifying or replacing the original harvested content.
- The application shall make provenance information accessible for inspection, verification, and export, subject to applicable access restrictions and output-format capabilities.
- Where provenance information is unavailable or cannot be established, the application shall explicitly record the limitation rather than generating unsupported provenance claims.

Verification criteria

- Execute predefined harvesting scenarios involving GBIF, OBIS, Figshare, and eLTER resources with different combinations of source identifiers, persistent identifiers, file structures, record identifiers, versions, and metadata availability.
- Verify that the application records provenance consistently at source, operation, and resource levels, and provides record- level and attribute-level traceability where required and technically feasible.
- Verify that original identifiers are preserved and remain distinguishable from internally generated identifiers.
- Perform multiple acquisitions of the same source dataset, including acquisitions with different retrieval parameters, timestamps, source versions, and acquisition outcomes.
- Verify that each acquisition receives a unique internal operation identifier, that previous acquisition histories are retained without modification, and that repeated acquisitions remain linked to their common source.
- Verify that harvested datasets, files, and identifiable records can be traced to their corresponding source resources and harvesting operations.
- Verify that the application does not create unsupported relationships between resources or records.
- Provide files with known sizes and checksums and verify that the application records and evaluates applicable integrity information.
- Simulate completed, incomplete, and failed harvesting operations and verify that provenance, acquisition status, relevant errors, and recovery information are retained.
- Execute scenarios involving deterministic relevance assessments, AI-generated recommendations, and explicit user approvals.
- Verify that the application records these activities as distinguishable events and correctly associates them with the affected resources or operations.

- Execute predefined Data Validation, Data Harmonisation, and Data Integration operations involving record corrections, taxonomic standardisation, attribute transformations, and dataset integration.
- Verify that the responsible modules record their operations using the common provenance model and that resulting outputs remain traceable to their original harvested inputs.
- Verify that original harvested values remain unchanged and that derived values, including record-level and attribute- level modifications where applicable, retain documented relationships to their source values and transformation operations.
- Execute data and metadata export scenarios and verify that relevant provenance information remains accessible and is included or referenced according to the capabilities and constraints of the selected export formats.
- Verify that unavailable provenance information is explicitly reported and that unsupported provenance claims are not generated.

RF07 | Harvesting Error and Response Handling

Functional requirements

- The application shall implement a common error handling, response management, and recovery framework for data harvesting operations across all supported sources, including GBIF, OBIS, Figshare, and eLTER.
- The application shall correctly handle paginated responses, empty results, asynchronous retrieval operations, API errors, rate limits, timeouts, interrupted transfers, inaccessible resources, and technical integrity failures.
- It shall distinguish valid empty responses from unsuccessful or incomplete acquisitions.
- The application shall classify retrieval errors according to their recoverability and automatically retry only operations affected by recoverable errors.
- Automatic retries shall follow documented policies defining retry limits, waiting intervals, exponential backoff where appropriate, and compliance with source-specific rate limits and server-provided retry instructions.
- Non- recoverable errors shall be reported without initiating inappropriate automatic retries.
- The application shall support checkpoint-based recovery and resumption of interrupted acquisitions where technically supported by the selected data access mechanism.
- Where safe resumption is unavailable, the application shall support re- execution through a new traceable harvesting operation, preserving the history and provenance of the interrupted acquisition.
- The application shall monitor the technical completeness and integrity of retrieval operations, including pagination completion, asynchronous job status, file transfer completion, and consistency between expected and retrieved outputs where determinable.
- It shall prevent incomplete or technically invalid outputs from being classified as successfully acquired resources.
- The application shall maintain acquisition status at both harvesting operation and individual resource levels, using consistent status definitions for In progress, Completed, Incomplete, and Failed.
- The overall operation status shall reflect the outcomes of its constituent retrieval activities without obscuring the individual status of each resource.
- The application shall preserve successfully retrieved data and technically usable partial outputs where possible, explicitly identifying incomplete resources and outstanding retrieval activities.
- Complete and technically verified resources shall remain eligible for Data Validation even when other independent resources within the same harvesting operation are incomplete or have failed.
- The application shall record errors, retry attempts, checkpoint information, recovery actions, acquisition statuses, and relevant diagnostic information through the common provenance framework defined by RF06.
- Recovery and re-execution activities shall remain traceable to their corresponding original harvesting operations without overwriting previous execution histories.

- The application shall provide users with clear information about retrieval progress, encountered errors, recovery attempts, incomplete resources, and available corrective actions.
- Error handling and recovery mechanisms shall not introduce scientific corrections, semantic harmonisation, or undocumented modifications to the harvested data.
- Functional requirement

Verification criteria

- Execute predefined harvesting scenarios across GBIF, OBIS, Figshare, and eLTER involving successful retrievals, paginated responses, valid empty results, asynchronous downloads, temporary service failures, rate limits, timeouts, authentication and authorisation errors, interrupted transfers, and technically invalid outputs.
- Verify that the application correctly distinguishes successful, incomplete, and failed retrieval operations and does not classify valid empty results as errors.
- Verify that pagination and asynchronous retrieval operations are considered complete only when the applicable completion conditions have been satisfied.
- Simulate recoverable errors, including temporary server failures, rate-limit responses, and connection timeouts.
- Verify that the application automatically retries eligible operations according to documented retry limits, waiting intervals, backoff policies, and server-provided instructions.
- Verify that non- recoverable errors do not trigger inappropriate automatic retries.
- Interrupt retrieval operations at predefined execution points and verify that the application preserves available checkpoint information and resumes acquisition when the selected mechanism supports safe continuation.
- Where resumption is unsupported, verify that recovery is performed through a new traceable harvesting operation without overwriting the original execution history.
- Provide test scenarios involving multiple resources within the same harvesting operation, including successfully acquired resources, incomplete downloads, inaccessible files, and failed retrievals.
- Verify that the application records resource-level statuses independently and assigns an overall operation status consistent with the constituent outcomes.
- Verify that complete and technically verified resources can proceed to Data Validation independently of incomplete or failed resources belonging to the same harvesting operation.
- Confirm that incomplete or technically invalid resources are not passed to Data Validation as completed acquisitions.
- Simulate corrupted files, missing pages, incomplete asynchronous downloads, and inconsistent retrieval counts.
- Verify that the application detects applicable integrity or completeness failures, preserves technically usable partial outputs where possible, and does not classify affected resources as successfully completed.
- Verify that error events, retry attempts, checkpoint information, recovery actions, execution timestamps, and final acquisition statuses are recorded through the common provenance framework.
- Confirm that repeated or recovered operations remain traceable without loss of previous execution histories.
- Verify that the application communicates retrieval progress, errors, recovery attempts, outstanding activities, and available corrective actions to users.
- Confirm that error handling and recovery do not alter the scientific content of harvested resources.
- Verification criterion Data Validation

Data Validation

RF08 | Structural data validation

Functional requirements

- The application shall validate the presence of required fields and the consistency of data types and formats according to documented validation rules.

Verification criteria

- Provide test datasets containing missing required fields, invalid data types, and malformed values.
- Verify that the application identifies and reports violations according to the applicable validation rules.

RF09 | Geographical coordinates validation

Functional requirements

- The application shall validate geographical coordinates by checking their format, completeness, and numerical ranges, and assess their consistency with available geographical information, including country, country code, and marine or terrestrial environment.

Verification criteria

- Provide test datasets containing valid and invalid coordinates, missing coordinate values, and inconsistencies between coordinates and associated geographical information.
- Verify that the application correctly identifies and reports invalid coordinates and potential geographical inconsistencies according to the applicable validation rules.

RF10 | Temporal data validation

Functional requirements

- The application shall validate temporal data by checking date formats, calendar validity, and consistency between temporal fields and the temporal coverage declared in dataset metadata, according to the applicable data schema or documented validation rules.

Verification criteria

- Provide test datasets containing valid and invalid dates, partial dates, inconsistent temporal fields, invalid date ranges, and records outside the declared temporal coverage.
- Verify that the application correctly identifies and reports temporal errors and potential inconsistencies according to the applicable validation rules.

RF11 | Duplicate records detection

Functional requirements

- The application shall identify exact and potential duplicate records within and across harvested datasets using record identifiers and predefined matching rules based on available data attributes, while preserving the original records and their provenance.

Verification criteria

- Provide test datasets containing identical records, records sharing the same source identifier, potentially duplicated records across different sources, and distinct observations with similar attributes.
- Verify that the application identifies and reports duplicate candidates according to the applicable matching rules without automatically removing or merging records.

RF12 | Identifier and relational integrity validation

Functional requirements

- The application shall validate the presence, format, and consistency of record identifiers and verify relationships between records when defined by the applicable data schema or dataset metadata.

Verification criteria

- Provide test datasets containing missing or malformed identifiers, inconsistent identifier usage, valid record relationships, and unresolved references between related records.
- Verify that the application identifies and reports identifier and relational integrity issues according to the applicable validation rules.

RF13 | Measurement and unit validation

Functional requirements

- The application shall validate measurement values and units by checking data types, unit validity, dimensional compatibility, and variable-specific constraints according to available metadata, controlled vocabularies, or predefined validation rules.

Verification criteria

- Provide test datasets containing valid and invalid measurement values, missing or unrecognised units, incompatible measurement units, and values violating predefined variable-specific constraints.
- Verify that the application correctly identifies and reports measurement- related errors and warnings according to the applicable validation rules.

RF14 | Schema identification and validation

Functional requirements

- The application shall identify declared data schemas or recognise potential schemas based on dataset structure and metadata, and apply the corresponding schema-specific validation rules once the schema has been confirmed.
- When the schema cannot be identified or is not supported, the application shall report this limitation and continue with applicable general validation checks.

Verification criteria

- Provide test datasets with declared and supported schemas, datasets with recognisable but undeclared schemas, datasets containing schema-specific violations, and datasets with unknown or unsupported schemas.
- Verify that the application correctly identifies or proposes the applicable schema, applies schema-specific validation rules only after confirmation, reports violations, and continues general validation when schema-specific checks are unavailable.

RF15 | Taxonomic validation

Functional requirements

- The application shall detect and flag taxonomic abbreviations, identification qualifiers, and nomenclatural annotations in scientific names using configurable recognition rules, preserving the original expressions for subsequent data harmonisation.
- The application shall validate distinct taxonomic queries using ChecklistBank (Catalogue of Life) and WoRMS TaxaMatch, retrieve available taxonomic information, assess taxonomic consistency, identify ambiguous or conflicting results, and associate all validation outcomes with the corresponding source records without modifying their original values.

Verification criteria

- Provide test datasets containing accepted names, synonyms, unmatched taxa, repeated scientific names, inconsistent taxonomic classifications, conflicting taxonomic matches, and scientific names containing abbreviations, identification qualifiers, and nomenclatural annotations.
- Verify that the application correctly detects and categorises these expressions, flags records requiring subsequent harmonisation, performs taxonomic matching without redundant requests for equivalent queries, reports taxonomic inconsistencies and conflicting results, and preserves the original scientific names and their associated annotations.

RF16 | Data validation report and interactive review

Functional requirements

- The application shall generate a structured validation report for each harvested dataset, summarising the outcomes of all applicable validation checks and providing record-level details of detected issues.
- The application shall provide an interactive interface for filtering, inspecting, and selectively reviewing validation issues, allowing users to record and track review decisions without requiring all issues to be reviewed before proceeding to data harmonisation.
- Validation results and user decisions shall be preserved separately, together with their provenance, without modifying the original data.

Verification criteria

- Provide test datasets containing valid records, multiple types of validation issues, non-applicable checks, and checks that cannot be completed.
- Verify that the application generates accurate dataset-level summaries and record-level validation results, supports filtering and inspection of issues, and allows users to record, update, and retrieve review decisions.
- Verify that reviewed and unreviewed issues remain distinguishable, that data harmonisation can proceed without mandatory review of all issues, and that validation results, user decisions, and original data are preserved with their provenance.
- Data Harmonisation

Data Harmonisation

RF17 | AI-assisted semantic variable matching

Functional requirements

- The application shall identify ble and propose semantic correspondences between variables across harvested datasets using AI-assisted matching supported by controlled vocabularies, ontologies, and available contextual metadata.
- The application shall assess candidate mappings based on variable definitions, measurement units, and relevant scientific context, and provide supporting semantic references and explanations.
- All proposed mappings shall require explicit user approval before being used for subsequent data harmonisation.
- The application shall preserve the proposed relationships, user decisions, and their provenance without modifying the original data.

Verification criteria

- Provide test datasets containing semantically equivalent, related, ambiguous, and non- equivalent variables with different labels, definitions, units, and contextual metadata.
- Verify that the application proposes and classifies candidate mappings using AI-assisted matching and configured semantic resources, provides supporting references and explanations, and allows users to

approve, reject, or revise the proposed relationships.

- Verify that unapproved mappings cannot be used for subsequent data harmonisation and that all proposals, decisions, and provenance information are preserved.

RF18 | AI-assisted adaptive Darwin Core mapping

Functional requirements

- The application shall map fields and data structures from heterogeneous source datasets to the Darwin Core standard, using predefined mapping rules and AI-assisted suggestions based on official Darwin Core term definitions, source metadata, sample values, and available semantic information.
- The application shall select an appropriate n Darwin Core representation according to the dataset content and supported data profiles, including Event, Occurrence, and MeasurementOrFact structures where applicable.
- New AI-generated mappings shall require explicit user approval before application.
- The application shall identify ambiguous or unsupported mappings, preserve unmapped source information, and document all applied mappings and their provenance.

Verification criteria

- Provide heterogeneous test datasets containing differently named fields corresponding to Darwin Core terms, event and occurrence information, measurements requiring structural mapping, ambiguous fields, and information without suitable Darwin Core correspondences.
- Verify that the application selects an appropriate supported Darwin Core representation, correctly applies predefined mappings, and generates AI-assisted mapping suggestions supported by Darwin Core definitions and source context.
- Verify that new AI-generated mappings require explicit user approval, that record relationships are preserved, that ambiguous and unsupported mappings are reported without forcing incorrect associations, and that unmapped source information and mapping provenance are retained.

RF19 | Transformation provenance and data lineage

Functional requirements

- The application shall preserve the original harvested data without modification and maintain structured provenance records for all data harmonisation operations.
- These records shall document the source and resulting data, original and harmonised values where applicable, applied transformations, mapping rules and their versions, d relevant user approvals, and execution outcomes.
- The application shall maintain traceable relationships between original and harmonised records, including cases involving structural transformations or multiple output records.
- Unsuccessful, skipped, and non-applicable transformations shall be documented with their corresponding reasons.

Verification criteria

- Provide test datasets requiring field mappings, unit conversions, semantic mappings, and structural transformations, including cases where one source record produces multiple harmonised records and where transformations fail, are skipped, or are not applicable.
- Verify that original data remain unchanged, that each transformation is documented with its input and output values where applicable, applied rules and versions, relevant user approvals, execution status, and reasons for unsuccessful or skipped operations.
- Verify that harmonised records and values can be traced to their original sources and that provenance information can be retrieved and inspected.

RF20 | AI-assisted measurement unit harmonisation

Functional requirements

- The application shall identify unit and harmonise measurement units across heterogeneous datasets using recognised unit definitions, dimensional compatibility checks, and deterministic conversion rules.
- The application shall support AI-assisted interpretation of non-standard or ambiguous unit representations, requiring explicit user approval for uncertain interpretations before conversion.
- Measurement values shall be converted to configured target units only when the conversion is scientifically valid and all required information is available.
- Original values and units shall be preserved, and all conversion operations and outcomes shall be documented through the transformation provenance mechanism.

Verification criteria

- Provide test datasets containing equivalent measurements expressed in different units, alternative unit representations, ambiguous unit labels, dimensionally incompatible units, and conversions requiring unavailable contextual information.
- Verify that the application correctly identifies units, checks dimensional compatibility, and performs accurate deterministic conversions to configured target units.
- Verify that uncertain AI-assisted interpretations require explicit user approval, that incompatible or unsupported conversions are not applied, and that original values, units, conversion outcomes, and user decisions are preserved and traceable.

RF21 | AI-assisted taxonomic harmonisation

Functional requirements

- The application shall harmonise taxonomic information across heterogeneous biodiversity datasets using taxonomic validation results, recognised taxonomic reference sources, and documented harmonisation rules.
- The application shall resolve supported synonym relationships, identify accepted taxon names and identifiers, and populate applicable Darwin Core taxonomic fields while preserving original identifications.
- The application shall recognise taxonomic abbreviations and identification qualifiers and, where appropriate, separate them into Darwin Core fields such as scientificName, identificationQualifier, and verbatimIdentification, without introducing unsupported taxonomic precision.
- AI- assisted interpretation shall be supported for ambiguous cases, requiring explicit user approval before applying uncertain resolutions.
- All harmonisation operations, reference sources, and user decisions shall be documented through the transformation provenance mechanism.

Verification criteria

- Provide test datasets containing accepted scientific names, synonyms, outdated names, incomplete identifications, taxonomic abbreviations, identification qualifiers, ambiguous matches, and names without reference matches.
- Verify that the application correctly reuses taxonomic validation results, resolves supported accepted names and synonym relationships, and populates applicable Darwin Core fields.
- Verify that qualifiers such as cf. and aff. are correctly represented in identificationQualifier, that original identifications are preserved in verbatimIdentification where appropriate, and that no unsupported taxonomic precision is introduced.
- Verify that ambiguous AI- assisted resolutions require explicit user approval and that all transformations, reference sources, and decisions are traceable.

RF22 | AI-assisted data format and value standardisation

Functional requirements

- The application shall standardise data formats and value representations across heterogeneous datasets according to the requirements of the target Darwin Core representation and documented standardisation rules.
- The application shall automatically apply deterministic transformations to recognised date formats, numeric representations, textual formats, and controlled values when their interpretation is unambiguous and scientifically valid.
- The application shall support AI-assisted interpretation of non-standard or ambiguous values using available source metadata and contextual information, requiring explicit user approval before applying uncertain interpretations.
- The application shall distinguish missing, unknown, non-applicable, unmeasured, and scientifically qualified values where such distinctions are supported by the source data, without introducing unsupported assumptions or altering their scientific meaning.
- Original values shall be preserved, unresolved cases shall be reported without forced transformation, and all standardisation operations, outcomes, and user decisions shall be documented through the transformation provenance mechanism.

Verification criteria

- Provide different date and numeric formats, inconsistent text, controlled value variants, ambiguous formats, missing markers, explicit zero and values below detection limits.
- Verify safe deterministic standardisation, approval of uncertain interpretations, preservation of meaningful distinctions, non-transformation of unresolved cases and provenance.
- Source: Harmonisation reference PDF, RF28 section

RF23 | Harmonisation review and decision management

Functional requirements

- a The application shall provide a ue unified interface for reviewing n and managing harmonisation proposals requiring user decisions across semantic variable matching, Darwin Core mapping, measurement unit harmonisation, taxonomic harmonisation, and data format and value standardisation.
- The application shall allow users to inspect proposed changes, supporting evidence, source data, and potential effects on the harmonised output, and to approve, reject, revise, or defer individual proposals.
- The application shall support applying approved decisions to multiple records when explicitly defined applicability conditions are satisfied.
- User decisions and their revision history shall be preserved with their provenance.
- Unapproved, rejected, or deferred proposals shall not be applied, while unaffected harmonisation operations shall be allowed to proceed.

Verification criteria

- Provide test datasets generating harmonisation proposals from multiple processes, including ambiguous semantic correspondences, uncertain Darwin Core mappings, measurement unit interpretations, taxonomic resolutions, and value standardisation cases.
- Verify that the application presents all proposals requiring user decisions in a unified interface, displays their supporting evidence and potential effects, and allows users to approve, reject, revise, or defer proposals.
- Verify that approved decisions can be applied to multiple records only when their applicability conditions are satisfied, that decision changes and histories are preserved, and that unapproved, rejected, or deferred proposals are not applied.

- Verify that unaffected harmonisation operations can proceed and that all decisions remain traceable to the corresponding proposals and source data.

RF24 | Comparative post-harmonisation quality control

Functional requirements

- The application shall perform comparative post- harmonisation quality control to verify the conformity of harmonised datasets with the selected Darwin Core representation, the integrity of their data structures and relationships, and the consistency of applied mappings and transformations with documented rules and user approvals.
- The application shall compare harmonised data with original datasets and pre- harmonisation validation results, using transformation provenance to distinguish pre- existing and introduced issues and to assess their resolution status.
- Detected issues shall be classified according to configurable severity rules, distinguishing critical errors, warnings, and informational findings.
- Datasets containing critical issues, or for which mandatory quality checks have not been completed, shall not be eligible for subsequent data integration.
- Non-critical issues shall be reported without preventing progression.
- The application shall generate a structured, non-destructive quality control report containing dataset-level summaries, record-level findings, relevant transformation references, and integration eligibility status.

Verification criteria

- Provide test datasets containing valid source records, pre-existing validation issues, correctly applied transformations, transformations introducing structural or semantic errors, unresolved harmonisation cases, and incomplete mandatory quality checks.
- Verify that the application checks conformity with the selected Darwin Core representation, structural and relational integrity, and consistency with approved mappings and documented transformation rules.
- Verify that the application correctly identifies issue origins and resolution statuses, assigns severity according to configured rules, and produces accurate dataset-level and record-level quality control reports with traceable transformation references.
- Verify that datasets with critical issues or incomplete mandatory checks are prevented from proceeding to data integration, while datasets containing only non-critical findings remain eligible.
- Verify that quality control does not modify original or harmonised data and that eligibility is reassessed after relevant corrections and checks.
- Data Integration

Data Integration

RF25 | Multi-source Darwin Core Dataset Integration

Functional requirements

- The application shall integrate records from multiple eligible and harmonised source datasets into a single logical Darwin Core dataset.
- The application shall support the integration of different but compatible Darwin Core data structures, including Event, Occurrence, and MeasurementOrFact where applicable, while preserving existing record relationships and avoiding the creation of unsupported entities or information.
- Source records shall remain distinct unless their reconciliation is supported by documented integration rules or explicitly approved user decisions.

- The integrated dataset shall maintain a coherent Darwin Core representation and preserve traceability to the contributing source datasets and records.

Verification criteria

- Provide two or more eligible and harmonised datasets containing different but compatible Darwin Core structures, including Event, Occurrence, and MeasurementOrFact records where applicable.
- Verify that the application produces a single coherent logical Darwin Core dataset while preserving valid relationships between records and without generating unsupported entities or values.
- Verify that distinct source records remain separate unless reconciliation is supported by documented rules or explicitly approved user decisions.
- Verify that the integrated dataset maintains traceability to all contributing source datasets and records, including cases where multiple source records are reconciled into a single integrated record.

RF26 | Integration Eligibility and Dataset Selection

Functional requirements

- The application shall allow users to select harmonised datasets for integration and determine their eligibility based on the results of the post-harmonisation quality control process.
- The application shall display each dataset's eligibility status, relevant quality findings, record counts, and Darwin Core structure.
- Datasets classified as blocked or not on assessed shall not be included in integration operations, while eligible datasets containing non-critical warnings may be selected without additional approval.
- The application shall perform preliminary structural compatibility and integration readiness checks to identify incompatible Darwin Core structures, missing integration prerequisites, and other conditions that may prevent integration.
- When a selection includes ineligible datasets, the application shall allow users to proceed with an eligible subset only after explicitly confirming the exclusions.
- Before integration, the application shall provide a summary of the selected datasets, excluded datasets, relevant quality findings, and compatibility issues.

Verification criteria

- Provide harmonised test datasets with different post-harmonisation quality control outcomes, including eligible datasets without issues, eligible datasets containing non- critical warnings, blocked datasets with critical issues, and datasets with incomplete mandatory quality checks.
- Include datasets with compatible and incompatible Darwin Core structures and different integration readiness conditions.
- Verify that the application correctly displays dataset eligibility, relevant quality findings, record counts, and Darwin Core structures.
- Verify that eligible datasets can be selected, including those with non-critical warnings without requiring additional approval, while blocked or not assessed datasets cannot be included in integration operations.
- Verify that preliminary structural compatibility and integration readiness checks identify incompatible structures and missing integration prerequisites.
- Verify that users can proceed with an eligible subset only after explicitly confirming the exclusion of ineligible datasets.
- Verify that the pre-integration summary accurately reports selected and excluded datasets, relevant quality findings, and compatibility issues, and that integration cannot proceed when unresolved blocking compatibility conditions remain.

RF27 | Cross-dataset Duplicate and Overlap Management

Functional requirements

- The application shall identify and manage duplicate records and overlapping information across harmonised datasets selected for integration, considering Event, Occurrence, and MeasurementOrFact structures where applicable.
- The application shall use deterministic matching rules based on reliable identifiers, source provenance, and documented record attributes, and support AI-assisted assessment of potential duplicates when deterministic evidence is insufficient.
- The application shall distinguish confirmed duplicates, potential duplicates, distinct observations, and partial overlaps, without treating similarities in taxonomic, spatial, or temporal attributes alone as sufficient evidence of record identity.
- Automatic reconciliation shall be permitted only when explicitly approved deterministic rules and their applicability conditions are satisfied.
- Ambiguous or AI-assisted reconciliation proposals shall require explicit user approval before application.
- The application shall preserve original source records, maintain traceable relationships between source and reconciled records, and document matching evidence, applied rules, reconciliation decisions, and unresolved cases.

Verification criteria

- Provide harmonised test datasets containing confirmed cross-source duplicates supported by reliable shared identifiers, potential duplicates with similar taxonomic, spatial, and temporal attributes, distinct observations sharing common characteristics, and partial overlaps involving Event, Occurrence, and MeasurementOrFact records.
- Verify that the application correctly identifies and classifies these cases using deterministic rules and AI-assisted assessment where appropriate.
- Verify that automatic reconciliation occurs only when explicitly approved deterministic rules and their applicability conditions are satisfied, and that ambiguous or AI- assisted proposals require explicit user approval.
- Verify that distinct observations are not incorrectly merged, that partial overlaps do not cause the loss of complementary information, and that unresolved cases remain identifiable without forced reconciliation.
- Verify that original source records are preserved and that matching evidence, applied rules, user decisions, reconciliation outcomes, and relationships between source and integrated records are recorded and traceable.

RF28 | Identifier and Relational Integrity Management

Functional requirements

- The application shall manage record identifiers and relationships during the integration of harmonised Darwin Core datasets, ensuring the unambiguous identification of integrated Event, Occurrence, and MeasurementOrFact records where applicable.
- The application shall preserve original source identifiers and assign distinct integrated identifiers when required to resolve identifier collisions or represent reconciled records.
- Reconciled records shall receive identifiers distinct from those of their contributing source records, while maintaining traceable links to all original identifiers.
- The application shall maintain documented mappings between source and integrated identifiers and consistently update dependent references, including eventID, parentEventID, occurrenceID, and other applicable relationships, using verified mappings.
- The application shall preserve existing valid relationships, prevent the creation of unsupported associations, and identify unresolved identifier conflicts or ambiguous relationships without applying speculative

corrections.

- All identifier assignments, mappings, and relationship updates shall be documented and traceable.

Verification criteria

- Provide harmonised test datasets containing identical identifier values referring to distinct records, valid and missing source identifiers, hierarchical Event relationships, Occurrence-to-Event references, MeasurementOrFact associations, and records reconciled according to approved integration rules or user decisions.
- Verify that the application correctly distinguishes identifier collisions from confirmed record identity, preserves original source identifiers, and assigns unambiguous integrated identifiers where required.
- Verify that reconciled records receive identifiers distinct from their contributing source identifiers and retain traceable links to all contributing records.
- Verify that dependent references are consistently updated through verified mappings, including hierarchical and cross-structure relationships.
- Verify that valid relationships are preserved, unsupported or ambiguous associations are not created automatically, and unresolved identifier or relationship conflicts are reported.
- Verify that all identifier assignments, source-to-integrated mappings, relationship updates, and relevant decisions are recorded and traceable.

RF29 | Cross-source Conflict Management

Functional requirements

- The application shall detect, classify, and manage conflicting or complementary attribute values across source records identified as representing the same entity during Darwin Core dataset integration, considering Event, Occurrence, and MeasurementOrFact records where applicable.
- The application shall distinguish consistent, complementary, conflicting, and unresolved values, taking into account their semantic compatibility, data context, and source provenance.
- The application shall support conflict resolution through explicitly approved rule-based procedures, AI-assisted recommendations, and manual user decisions.
- Automatic resolution shall be permitted only when documented and approved rules and their applicability conditions are satisfied.
- AI- assisted recommendations and ambiguous resolutions shall require explicit user approval before application.
- Complementary information may be integrated through approved rules only when semantic compatibility is established and the resulting representation does not introduce unsupported information or associations.
- The application shall distinguish blocking from non- blocking conflicts according to their impact on the scientific and structural validity of the integrated representation.
- Unresolved blocking conflicts shall prevent only the affected reconciliation, allowing independent integration operations to proceed.
- Non- blocking conflicts may remain unresolved, provided that the discrepancies are explicitly documented and no unsupported value is selected.
- Original source values shall be preserved, and all applied resolutions, supporting evidence, decisions, and unresolved conflicts shall remain traceable.

Verification criteria

- Provide harmonised test datasets containing source records confirmed as representing the same entity, with consistent values, semantically compatible complementary information, incompatible complementary information, conflicting attribute values, and unresolved discrepancies across Event, Occurrence, and MeasurementOrFact records.

- Verify that the application correctly classifies these cases and applies automatic conflict resolution or complementary information integration only when explicitly approved rules and their applicability conditions are satisfied.
- Verify that semantically incompatible information is not automatically combined and that AI- assisted recommendations provide supporting evidence and require explicit user approval before application.
- Verify that users can review conflicting source values, approve or reject proposed resolutions, select a justified alternative, or retain conflicts as unresolved.
- Verify that the application distinguishes blocking and non-blocking conflicts based on their impact on the validity of the integrated representation.
- Verify that unresolved blocking conflicts prevent only the affected reconciliation without interrupting independent integration operations, while unresolved non-blocking conflicts remain explicitly documented without arbitrary value selection.
- Verify that original source values and provenance are preserved and that applied rules, supporting evidence, user decisions, resolution outcomes, and unresolved conflicts are recorded and traceable.

RF30 | Integrated Dataset Provenance

Functional requirements

- The application shall capture and maintain structured provenance information for the integration of harmonised Darwin Core datasets, ensuring traceability from the integrated dataset and its individual records to their contributing source datasets and records.
- Provenance shall be maintained at dataset and record levels, with attribute- level provenance for values selected, combined, or reconciled across sources.
- The application shall document integration operations, including cross- dataset duplicate identification, record reconciliation, identifier assignment and remapping, relationship updates, complementary information integration, conflict resolution, and the handling of unresolved cases.
- Provenance records shall include the applicable rules, supporting evidence, AI- assisted recommendations where relevant, user approvals or rejections, and the outcomes of integration decisions.
- The application shall maintain a structured provenance record separate from, but explicitly linked to, the integrated Darwin Core dataset.
- This record shall preserve source-to-integrated identifier mappings and provide connections to provenance information generated during previous data processing stages without unnecessary duplication.
- The application shall record relevant integration execution information, including references to input datasets and their versions or acquisition states, applied integration configurations and rules, execution identifiers, and output dataset versions.
- The recorded provenance shall support inspection, auditing, traceability, and reproducibility of integration operations to the extent permitted by the availability of the original inputs and execution dependencies.

Verification criteria

- Provide harmonised test datasets with documented source provenance and execute integration scenarios involving distinct records, confirmed cross- source duplicates, reconciled records, identifier collisions and remapping, relationship updates, complementary information, conflicting attribute values, AI-assisted recommendations, user decisions, and unresolved cases.
- Verify that the application records the contributing source datasets and records for each integrated record and preserves traceable mappings between original and integrated identifiers.
- Verify that attribute-level provenance is available for values selected, combined, or reconciled across sources, allowing the contributing source values and the rationale for the resulting values to be identified.
- Verify that integration operations, applied rules, supporting evidence, AI- assisted recommendations, user approvals or rejections, reconciliation outcomes, and unresolved cases are documented and retrievable.

- Verify that provenance information is maintained in a structured record separate from, but explicitly linked to, the integrated Darwin Core dataset, and that connections to provenance records from previous processing stages are preserved.
- Verify that provenance records contain the relevant integration execution identifiers, input dataset references and versions or acquisition states, applied configurations and rules, and output dataset versions.
- Verify that users can reconstruct the integration history of selected records and inspect the decisions contributing to their final representation.
- Where the original inputs and execution dependencies are available, verify that the recorded information supports reproduction or independent verification of the applied integration operations and decisions.

RF31 | Post-integration Quality Control

Functional requirements

- The application shall perform comparative, non-destructive quality control on the integrated Darwin Core dataset to verify the correctness and consistency of integration operations, using the harmonised source datasets and their existing quality control results as reference.n The application shall assess Darwin Core structural integrity, identifier uniqueness and relational consistency, duplicate and reconciliation outcomes, conflict resolution consistency, preservation of source information, and completeness and consistency of integration provenance.
- The application shall identify unexpected record or attribute loss, unsupported entities or relationships, incorrect identifier mappings, unjustified reconciliations, unresolved critical conflicts affecting integrated records, and missing or inconsistent provenance information.
- The application shall classify identified issues according to their severity, associate them with the affected records, attributes, relationships, or integration operations, and generate a structured quality report containing supporting evidence and relevant corrective recommendations.
- The application may provide AI-assisted recommendations for interpreting and addressing identified issues, without automatically modifying the integrated dataset.
- The application shall assign the integrated dataset an eligibility status of eligible, blocked, or not assessed.
- Critical errors or incomplete mandatory checks shall prevent the dataset from being considered eligible for subsequent analytical data preparation, while documented non-critical warnings shall not automatically prevent progression.
- Quality control results and subsequent verification outcomes shall remain traceable.

Verification criteria

- Provide integrated Darwin Core test datasets generated from harmonised sources, including valid integration outcomes and controlled cases involving identifier collisions, broken relationships, incorrect reconciliation, unresolved conflicts, unexpected record or attribute loss, unsupported associations, and incomplete provenance records.
- Verify that the application performs the required structural, relational, reconciliation, conflict management, preservation, and provenance checks, comparing integration results with the source datasets and existing quality control findings where relevant.
- Verify that valid reconciliations and documented changes in record counts are not incorrectly classified as data loss, while unsupported or unjustified changes are identified.
- Verify that each detected issue is classified by severity and linked to the affected records, attributes, relationships, or integration operations, with supporting evidence and relevant corrective recommendations.
- Verify that the quality control process does not automatically modify the integrated dataset and that AI-assisted recommendations, where provided, remain advisory.
- Verify that the application assigns eligible status when mandatory checks are completed without critical errors, including cases with documented non-critical warnings; blocked status when critical errors are

detected; and not assessed status when mandatory checks are incomplete.

- Verify that blocked or not assessed datasets cannot progress to analytical data preparation and that quality control findings and subsequent verification outcomes are recorded and traceable.
- Data Export

Data Export

RF32 | Data export

Functional requirements

- The application shall allow users to export integrated Darwin Core datasets in Darwin Core Archive (DwC-A) format as the primary export format and CSV as an alternative tabular format.
- The application shall support the export of either the complete integrated dataset or selected data components, including Event, Occurrence, and MeasurementOrFact records where applicable.
- The application shall preserve the relevant Darwin Core structure, integrated identifiers, attribute values, and valid relationships in the exported data.
- When exporting selected components, the application shall identify and manage dependencies between records, ensuring that the export does not introduce invalid relationships or misleading representations of the original integrated dataset.
- CSV exports shall retain the applicable Darwin Core terms and provide a clear representation of the exported components without forcing incompatible relational structures into a single flat table.
- The application shall allow the export of datasets classified as Eligible, Blocked, or Not assessed according to the post-integration quality control process.
- For datasets classified as Blocked or Not assessed, the application shall explicitly communicate their quality status and associated limitations before export.
- Relevant quality information and warnings shall accompany the exported data, allowing users to distinguish datasets that have passed mandatory quality checks from those requiring further assessment or correction.
- The application shall maintain consistency between the exported data and the corresponding version of the integrated dataset, including relevant dataset identifiers, version information, and references to the associated integration provenance.
- Export operations shall not modify the integrated dataset or its provenance records.

Verification criteria

- Provide integrated Darwin Core test datasets containing Event, Occurrence, and MeasurementOrFact records, including datasets with hierarchical Event relationships, cross- component references, reconciled records, and documented integration provenance.
- Include datasets classified as Eligible, Blocked, and Not assessed.
- Verify that the application successfully exports complete datasets in Darwin Core Archive (DwC- A) format and supports CSV exports of the applicable data components.
- Verify that exported data preserve the relevant Darwin Core terms, attribute values, integrated identifiers, and valid record relationships, and that the generated DwC-A packages conform to the applicable archive structure and metadata requirements.
- Verify that users can select individual data components for export and that the application identifies and appropriately handles dependencies between selected and related records.
- Verify that partial exports do not introduce invalid references, unsupported relationships, or misleading representations of the original dataset, and that CSV exports preserve the relevant structure and meaning of the exported information.
- Verify that datasets classified as Eligible, Blocked, or Not assessed can be exported.

- Verify that the application explicitly displays the quality status and relevant warnings or limitations before exporting Blocked or Not assessed datasets, and that the exported output includes or is accompanied by the corresponding quality information.
- Verify that exported datasets retain consistent dataset identifiers, version references, and links to the associated integration provenance.
- Verify that the export process does not modify the original integrated dataset, its identifiers, relationships, or provenance records.

RF33 | Metadata export

Functional requirements

- The application shall support the generation, validation, and export of structured metadata describing integrated Darwin Core datasets, using Ecological Metadata Language (EML) as the supported metadata export standard.
- The application shall generate integrated-dataset metadata through deterministic extraction and derivation of rt relevant information from source metadata, integrated records, and documented previous processing stages.
- The generated metadata shall include, where available and applicable, dataset identification and description, geographic, temporal and taxonomic coverage, methodological information, source attribution, rights and licensing information, data quality status, and references to integration provenance.
- The application shall distinguish metadata describing individual source datasets from metadata describing the integrated dataset and shall preserve the provenance of information used in metadata generation.
- The application shall support AI-assisted recommendations for descriptive metadata and manual user review and approval of generated, ambiguous, conflicting, or non-derivable information.
- AI- generated recommendations shall not be applied without explicit user approval, and the application shall not introduce unsupported scientific, methodological, attribution, or licensing information.
- The application shall identify missing, conflicting, or inconsistent metadata and validate the generated EML against the applicable structural and mandatory metadata requirements.
- The application shall permit the export of incomplete or non- compliant EML metadata, provided that the user is explicitly informed of the identified issues and that the exported metadata are accompanied by a validation report clearly distinguishing compliant from non-compliant outputs.
- Non-compliant metadata shall not be presented as valid EML.
- The application shall ensure that exported metadata accurately describe the corresponding complete or partial data export, including its content, coverage, identifiers, version, provenance references, and relevant quality limitations.
- Metadata export shall not modify the underlying integrated Darwin Core dataset or its provenance records.

Verification criteria

- Provide integrated Darwin Core test datasets with associated source metadata, including complete, incomplete, conflicting, and source- specific metadata, documented integration provenance, and different post-integration quality statuses.
- Include scenarios involving complete and partial data exports.
- Verify that the application generates and exports EML metadata through deterministic extraction and derivation of supported information from source metadata, integrated records, and previous processing stages.
- Verify that dataset identification, description, geographic, temporal and taxonomic coverage, methodology, attribution, rights and licensing information, quality status, and provenance references are correctly represented where available and applicable, without introducing unsupported information.

- Verify that the application distinguishes source- specific metadata from integrated-dataset metadata, identifies missing or conflicting information, and supports manual review and approval.
- Verify that AI- assisted recommendations are presented with appropriate contextual information and require explicit user approval before incorporation into the exported metadata.
- Verify that unsupported scientific, methodological, attribution, or licensing information is not automatically generated or included.
- Verify that the application validates generated EML against the applicable structural and mandatory metadata requirements and correctly identifies missing mandatory elements, structural errors, and other compliance issues.
- Verify that compliant metadata are clearly identified as valid and that incomplete or non-compliant metadata can still be exported with explicit user warnings and an accompanying validation report documenting the detected issues.
- Verify that non-compliant outputs are never incorrectly labelled as valid EML.
- Verify that metadata associated with complete or partial data exports accurately describe the corresponding exported content, including consistent dataset identifiers, version references, geographic, temporal and taxonomic coverage, provenance links, and relevant quality limitations.
- Verify that metadata export does not modify the integrated dataset or its provenance records.
- Data Analysis

Data Analysis

RF34 | Exploratory and Ecological Data Analysis

Functional requirements

- The application shall provide configurable exploratory and ecological data analysis functionalities to support the examination, description, visualisation, and comparison of harmonised and integrated biodiversity datasets.
- The application shall provide a catalogue of supported d analytical methods, including a dataset exploration, descriptive statistics, data aggregation, taxonomic composition, observed taxonomic richness, abundance-based summaries, spatial distributions, temporal patterns, and descriptive ecological comparisons.
- Users shall be able to select analytical methods and configure their parameters, including variables, filters, grouping criteria, aggregation methods, and visualisation options, within the constraints of each supported method.
- Before executing an analysis, the application shall assess the availability and suitability of the required data, including relevant variables, measurement units, taxonomic resolution, temporal and spatial information, and sampling context where applicable.
- The application shall prevent analyses that cannot be performed with the available data and explicitly communicate limitations that may affect the validity or interpretation of otherwise executable analyses.
- All analytical operations shall be performed on selected, versioned datasets without modifying the underlying data.
- The application shall preserve analytical configurations, input dataset references, method identifiers and versions, parameters, execution outcomes, and results to support reproducibility, export, and integration with the reporting module.
- Detailed Functional Specifications
- RF34.1 – Dataset Exploration The application shall display dataset structure, variables, data types, record counts, missing values, and available metadata.

- RF34.2 – Data Filtering and Selection The application shall allow users to select records through taxonomic, geographical, temporal, environmental, and attribute- based filters.
- RF34.3 – Descriptive Statistics The application shall calculate appropriate descriptive statistics for numerical and categorical variables, including counts, frequencies, mean, median, standard deviation, minimum, and maximum.
- RF34.4 – Data Aggregation and Grouping The application shall aggregate records according to user-selected variables, observational units, grouping criteria, and supported aggregation methods.
- RF34.5 – Taxonomic Composition Analysis The application shall describe the distribution of observations across taxonomic groups and selected taxonomic ranks.
- RF34.6 – Observed Taxonomic Richness The application shall calculate the number of distinct observed taxa within selected groups, considering taxonomic resolution and identification consistency.
- RF34.7 – Abundance-Based Analysis The application shall calculate abundance summaries and distributions when suitable abundance measurements, units, and observational context are available.
- RF34.8 – Spatial Distribution Analysis The application shall support the exploration and visualisation of the geographical distribution of observations and selected variables.
- RF34.9 – Temporal Pattern Analysis The application shall describe temporal distributions and changes in observed variables using configurable temporal groupings.
- RF34.10 – Ecological Comparisons The application shall support descriptive comparisons of taxonomic composition, observed richness, abundance, and other available measurements across sites, periods, habitats, or environmental conditions.
- RF34.11 – Data Visualisation The application shall generate configurable charts, tables, and maps appropriate to the selected variables and analytical results.
- RF34.12 – Analytical Method Catalogue and Configuration The application shall provide a catalogue of supported analytical methods, allowing users to configure valid parameters and analysis options. Each method shall specify its required data, supported parameters, applicability conditions, methodological limitations, and available output types.
- RF34.13 – Analysis Applicability Assessment The application shall evaluate data requirements, measurement compatibility, observational units, and relevant methodological limitations before analysis execution.
- RF34.14 – Analysis Provenance and Reproducibility The application shall record the input dataset identifiers and versions, selected records or selection criteria, analytical method identifiers and versions, configuration parameters, aggregation rules, execution status, generated results, and identified limitations. The recorded information shall support the reproducibility of analytical operations and the identification of the exact dataset and method versions used.
- RF34.15 – Analysis Results Export The application shall allow users to export individual analytical results, including tabular outputs and visualisations, together with the relevant configuration, units, and interpretative limitations.
- RF34.16 – Reporting Integration The application shall make selected analytical results available to RF35 – Progressive and Final Report Generation. The information provided shall include references to the input dataset versions, analytical methods and versions, configuration parameters, generated results, execution outcomes, and identified limitations. The reporting module shall consume these recorded outputs without independently recalculating analytical results.

Verification criteria

- VC34.1 – Dataset Exploration Load a versioned integrated dataset and verify that the application correctly displays its structure, variables, data types, record counts, missing values, and available metadata.
- VC34.2 – Data Filtering and Selection Apply taxonomic, spatial, temporal, and attribute- based filters. Verify that the selected records correspond to the specified conditions and that the original dataset remains unchanged.

- VC34.3 – Descriptive Statistics Calculate descriptive statistics for numerical and categorical variables and compare the results with independently calculated reference values, including cases with missing data.
- VC34.4 – Data Aggregation Group records using different observational units and aggregation criteria. Verify that the resulting counts and aggregated values correspond to the selected configuration.
- VC34.5 – Taxonomic Composition Analyse the taxonomic composition of a dataset and verify that taxonomic group assignments and record counts are consistent with the available taxonomic information.
- VC34.6 – Observed Taxonomic Richness Calculate observed taxonomic richness at different taxonomic resolutions. Verify that distinct taxa are counted consistently and that relevant taxonomic ambiguities are identified.
- VC34.7 – Abundance- Based Analysis Execute abundance summaries using suitable quantitative data. Verify the correct interpretation of values and units and the identification of missing or incompatible abundance information.
- VC34.8 – Spatial Distribution Analysis Generate spatial representations using datasets containing geographical information. Verify that plotted locations correspond to the selected records and that invalid or missing coordinates are handled appropriately.
- VC34.9 – Temporal Pattern Analysis Generate temporal summaries using configurable time intervals. Verify correct temporal grouping and handling of missing or incomplete dates.
- VC34.10 – Ecological Comparisons Compare selected sites, periods, or habitats. Verify that the application identifies relevant differences in measurement units, sampling methods, or observational units that may affect comparability.
- VC34.11 – Data Visualisation Generate charts, tables, and maps from analytical results. Verify that displayed values, labels, units, and grouping criteria correspond to the calculated outputs.
- VC34.12 – Analytical Method Catalogue Verify that supported analytical methods are available through the catalogue and that each method provides its description, required data, configurable parameters, applicability conditions, limitations, and supported outputs.
- VC34.13 – Method Configuration Configure different analyses using valid combinations of variables, filters, grouping criteria, and supported parameters. Verify that invalid configurations are rejected or require correction before execution.
- VC34.14 – Applicability Assessment Attempt to execute analyses using datasets with missing required variables, incompatible measurements, or insufficient contextual information. Verify that the application correctly identifies applicable, applicable-with-limitations, and non-applicable cases, preventing execution when essential requirements are not satisfied.
- VC34.15 – Reproducibility and Method Versioning Repeat an analysis using the same dataset version, method version, filters, and parameters. Verify that the results are reproducible and that all relevant configuration and provenance information is preserved. Verify that results generated using different method versions can be distinguished and traced to their respective implementations.
- VC34.16 – Results Export Export analytical tables and visualisations. Verify that exported outputs are consistent with the displayed results and include the relevant analytical configuration and limitations.
- VC34.17 – Reporting Integration Make selected analytical results available to RF35. Verify that the reporting module receives the correct input dataset versions, analytical method identifiers and versions, parameters, execution outcomes, results, and limitations. Verify that RF35 presents the recorded analytical results without independently recalculating them.
- VC34.18 – Non- Destructive Analysis Execute and repeat analytical operations. Verify that the original datasets, integrated datasets, and previously recorded processing decisions remain unchanged. Reporting

Reporting

RF35 | Progressive and Final Report Generation

Functional requirements

- The application shall progressively compile a structured report documenting the execution of the data processing workflow, including data harvesting, validation, harmonisation, integration, analysis, and export.
- The report shall be generated from structured processing results, provenance records, recorded user decisions, and analytical outputs produced by the corresponding application modules.
- The application shall provide an interactive interface for reviewing the report, identifying incomplete ord outdated sections, and examining the results and decisions associated with each workflow stage.
- Upon completion of the workflow, the application shall allow users to generate a consolidated report representing a consistent snapshot of the available processing records and selected analytical results.
- The final report shall be exportable in PDF and JSON formats.
- Each generated report snapshot shall be versioned and preserved without modifying the underlying datasets, analytical results, or recorded workflow decisions.
- Detailed Functional Specifications
- RF35.1 – Progressive Report Compilation The application shall progressively compile the report as processing stages are executed, including Data Harvesting, Data Validation, Data Harmonisation, Data Integration, Data Analysis, and Data Export. Report sections shall be updated using the structured records produced by the corresponding modules.
- RF35.2 – Workflow Overview The report shall provide an overview of the workflow execution, including its identifier, selected input datasets, execution stages, processing status, and relevant timestamps.
- RF35.3 – Input Datasets and Harvesting The report shall document input dataset identifiers, sources, acquisition operations, retrieval outcomes, and relevant harvesting limitations.
- RF35.4 – Data Validation The report shall document validation operations, identified issues, affected records, validation outcomes, and relevant user decisions.
- RF35.5 – Data Harmonisation The report shall document schema mappings, semantic harmonisation decisions, transformations, unit conversions, taxonomic operations, and associated limitations.
- RF35.6 – Data Integration and Export The report shall document dataset integration operations, duplicate and conflict management, identifier handling, integrated dataset versions, export operations, and generated output references.
- RF35.7 – Data Analysis The report shall provide a dedicated Data Analysis section containing references to the analytical operations performed through RF34. For each selected analytical result, the report shall include the analytical method and version, input dataset version, relevant filters and configuration parameters, execution outcome, generated results, and identified limitations. The report may include selected analytical tables and visualisations or references to their preserved outputs. The reporting module shall not independently recalculate analytical results.
- RF35.8 – Provenance and Processing History The report shall preserve references to the processing operations, source datasets, transformations, analytical executions, recorded decisions, and associated provenance information.
- RF35.9 – Report Completeness and Status Management The application shall identify the completion status of each report section using the following states: • Not started • In progress • Completed • Completed with warnings • Failed • Skipped • Outdated The application shall identify report sections requiring updates following the re- execution of relevant workflow operations.
- RF35.10 – Interactive Report Consultation The application shall allow users to navigate report sections, inspect processing outcomes, review recorded decisions, and examine associated analytical results and limitations. The interactive report shall display verified processing information independently of any optional AI-generated explanations.

- RF35.11 – Final Report Consolidation and Export The application shall generate a consolidated report snapshot representing a consistent state of the workflow execution. The final report shall be exportable in PDF and JSON formats. These exports shall document the overall workflow and shall remain distinct from the individual analytical result exports provided by RF34. Both formats shall represent the same report snapshot and preserve the relevant processing information, analytical results, provenance references, and limitations.
- RF35.12 – Report Versioning and Snapshot Preservation The application shall assign an identifier and version to each consolidated report snapshot. Previously generated snapshots shall remain immutable and accessible, even when the underlying workflow is subsequently re- executed or updated. New processing results shall require the generation of a new report snapshot rather than modification of an existing one.

Verification criteria

- VC35.1 – Progressive Compilation Execute the workflow incrementally and verify that the report is progressively populated using records from harvesting, validation, harmonisation, integration, analysis, and export.
- VC35.2 – Workflow Overview Verify that the report correctly identifies the workflow execution, input datasets, processing stages, execution status, and relevant timestamps.
- VC35.3 – Harvesting Documentation Execute harvesting operations with successful and unsuccessful retrieval outcomes. Verify that sources, acquired datasets, execution results, and limitations are accurately documented.
- VC35.4 – Validation Documentation Execute validation checks producing different outcomes. Verify that identified issues, affected records, validation results, and recorded user decisions are correctly represented.
- VC35.5 – Harmonisation Documentation Execute schema mappings, semantic harmonisation, and data transformations. Verify that the report accurately documents the operations, parameters, decisions, and limitations.
- VC35.6 – Integration and Export Documentation Execute integration and export operations. Verify that the report correctly records integrated dataset references, identifier handling, duplicate and conflict management, export formats, and generated outputs.
- VC35.7 – Data Analysis Documentation Execute multiple analytical methods through RF34 and select results for inclusion in the report. Verify that the Data Analysis section correctly identifies the input dataset versions, analytical methods and versions, filters, parameters, execution outcomes, results, and limitations. Verify that the report presents the recorded analytical outputs without independently recalculating them.
- VC35.8 – Provenance Consistency Compare the information displayed in the report with the underlying processing, analytical, and provenance records. Verify that references, decisions, and execution outcomes remain consistent.
- VC35.9 – Section Status Management Execute, skip, fail, and re- execute different workflow stages. Verify that report sections correctly transition between the defined status values and that dependent sections are marked as outdated when appropriate.
- VC35.10 – Interactive Consultation Navigate the report and inspect different processing stages, recorded decisions, analytical outputs, and limitations. Verify that displayed information corresponds to the underlying records.
- VC35.11 – Final Report Consolidation Generate a final report snapshot and verify that it represents a consistent state of the workflow, including the analytical results selected for inclusion.
- VC35.12 – PDF and JSON Export Consistency Export the same report snapshot in PDF and JSON formats. Verify that both outputs represent the same report version and contain consistent processing information, analytical results, provenance references, and limitations.
- VC35.13 – Report Versioning Generate successive report snapshots after updating or re-executing workflow operations. Verify that each snapshot receives a distinct identifier or version and can be retrieved independently.
- VC35.14 – Snapshot Immutability Generate a report snapshot and subsequently modify the workflow through an authorised processing operation. Verify that the previously generated snapshot remains unchanged and

that updated results appear only in a newly generated snapshot.

- VC35.15 – Non- Destructive Reporting Generate, consult, and export reports. Verify that reporting operations do not modify source datasets, integrated datasets, analytical outputs, or recorded user decisions. AI Assistance

AI Assistance

RF36 | Context-Aware AI Assistance with User-Approved Actions

Functional requirements

- The application shall provide a context-aware AI assistant accessible throughout the data processing workflow, including data harvesting, validation, harmonisation, integration, analysis, and reporting.
- The assistant shall support both on-demand natural- language interaction and proactive, non-intrusive contextual suggestions.
- The assistant shall use relevant information from the current workflow context, structured processing and analytical results, provenance records, recorded user decisions, and available reference documentation.
- The assistant shall provide evidence-grounded explanations, contextual recommendations, and support for configuring operations available through the application modules.
- Where supported by the corresponding modules, the assistant shall prepare structured processing or analytical actions for user review.
- Before execution, the application shall present the proposed action, its parameters, affected data, and expected effects for explicit user approval.
- Approved actions shall be executed exclusively through the corresponding application modules, subject to their validation, authorisation, and provenance mechanisms.
- The assistant shall not independently modify datasets, processing outputs, analytical results, workflow configurations, or recorded user decisions.
- The application shall maintain traceability of AI-assisted actions and user decisions, manage uncertainty and execution errors, and ensure that proactive suggestions do not interrupt or automatically alter the workflow.
- Detailed Functional Specifications
- RF36.1 – Natural-Language Interaction The application shall allow users to submit questions, requests, and instructions through a conversational interface.
- RF36.2 – Workflow Context Awareness The assistant shall identify the active workflow, selected datasets, current processing stage, execution status, and available processing or analytical results.
- RF36.3 – Contextual Information Retrieval The assistant shall retrieve relevant structured processing results, analytical outputs, provenance records, user decisions, and reference documentation, subject to the user's access permissions.
- RF36.4 – Processing and Analytical Results Explanation The assistant shall explain validation issues, harmonisation operations, integration decisions, analytical results, and workflow outcomes using available evidence. The assistant shall distinguish verified results from AI- generated interpretations.
- RF36.5 – Contextual Recommendations The assistant shall suggest possible actions for resolving processing issues, improving data quality, configuring supported analyses, or completing workflow operations. Recommendations shall not be automatically applied.
- RF36.6 – AI-Assisted Action and Analysis Configuration Preparation The assistant shall prepare structured actions compatible with the functionalities of the corresponding application modules. For Data Analysis, the assistant shall support the selection of analytical methods from the RF34 catalogue and the preparation of valid configurations, including variables, filters, grouping criteria, aggregation methods, and other supported parameters. The assistant shall not independently generate or execute arbitrary analytical code outside the supported method catalogue.

- RF36.7 – Action Preview and User Approval The application shall present proposed actions, configuration parameters, affected data, expected effects, and available alternatives before requesting explicit user approval.
- RF36.8 – Controlled Action Execution Approved actions shall be executed through the corresponding application modules, respecting their validation, applicability, authorisation, and provenance mechanisms. The application shall revalidate proposals before execution and request renewed approval when relevant data or parameters have changed.
- RF36.9 – Evidence and Provenance References The assistant shall associate factual explanations and recommendations with relevant processing records, analytical results, or supporting documentation.
- RF36.10 – Uncertainty and Error Management The assistant shall communicate missing information, ambiguous interpretations, unsupported requests, methodological limitations, and actual execution failures.
- RF36.11 – AI Interaction and Decision Traceability The application shall record relevant action proposals, user approvals or rejections, execution outcomes, and supporting references when they affect workflow decisions.
- RF36.12 – AI-Assisted Reporting The assistant shall provide natural-language explanations and summaries based on structured reporting information produced by RF35. Each report-related explanation or summary shall be associated with the specific report version or snapshot used as its source. AI-generated explanations shall be distinguishable from verified report contents and shall not modify the underlying report or snapshot.
- RF36.13 – Proactive Contextual Suggestions The assistant shall present relevant, non-intrusive contextual recommendations when workflow results indicate issues or opportunities for user intervention. For Data Analysis, the assistant may suggest supported analytical methods when the available data satisfy the corresponding applicability requirements. Proactive suggestions shall not automatically initiate or execute processing or analytical operations.
- RF36.14 – Suggestion Management and User Control The application shall allow users to dismiss or disable proactive suggestions. Repeated suggestions for unchanged issues shall be avoided, and dismissing suggestions shall not affect workflow execution or recorded processing decisions.

Verification criteria

- VC36.1 – Natural- Language Interaction Verify that users can submit natural-language questions and requests and receive responses relevant to the current workflow context.
- VC36.2 – Workflow Context Awareness Execute different workflow stages, including Data Analysis, and verify that the assistant correctly identifies the relevant datasets, active stage, execution status, and available results.
- VC36.3 – Contextual Information Retrieval Submit requests requiring information from multiple workflow stages. Verify that the assistant retrieves the appropriate processing records, analytical outputs, user decisions, and supporting documentation while respecting access permissions.
- VC36.4 – Evidence- Grounded Explanations Request explanations of validation issues, harmonisation decisions, integration outcomes, and analytical results. Verify that factual statements are consistent with the underlying records and that AI-generated interpretations are distinguishable from verified results.
- VC36.5 – Contextual Recommendations Present documented processing issues or analytical requirements and verify that the assistant proposes relevant actions without automatically applying changes.
- VC36.6 – AI-Assisted Analysis Configuration Request an ecological analysis through natural language. Verify that the assistant selects a supported method from the RF34 catalogue and prepares a structured configuration containing the required variables, filters, grouping criteria, and parameters. Verify that unsupported methods or invalid parameter combinations are not presented as executable configurations.
- VC36.7 – Action Preview and Approval Verify that proposed processing and analytical actions are presented with their expected effects and that users can approve, modify, or reject them before execution.
- VC36.8 – Controlled Execution Approve a proposed action and verify that it is executed exclusively through the corresponding application module, subject to its validation, applicability, and authorisation mechanisms. Verify that rejected actions are not executed.

- VC36.9 – Changed Context Handling Prepare an action and modify the relevant workflow state before execution. Verify that the proposal is revalidated and that renewed approval is requested when the previously approved parameters or affected data have changed.
- VC36.10 – Uncertainty and Error Management Submit requests involving missing, ambiguous, or unsupported information and verify that the assistant communicates the limitations. Simulate an execution failure and verify that the actual outcome is reported accurately.
- VC36.11 – Interaction and Decision Traceability Verify that relevant action proposals, approvals, rejections, execution outcomes, and supporting references are recorded and associated with the corresponding workflow execution.
- VC36.12 – Report Version Grounding Generate two different RF35 report snapshots and request an AI-generated explanation of each. Verify that the assistant uses the correct report version, identifies its source, and does not combine results from different snapshots without explicitly distinguishing them.
- VC36.13 – Proactive Contextual Suggestions Trigger relevant workflow events, including validation issues, integration conflicts, and the availability of data suitable for supported ecological analyses. Verify that the assistant presents appropriate contextual suggestions without requiring a direct user request or interrupting the workflow.
- VC36.14 – Suggestion Management Verify that users can dismiss or disable proactive suggestions, that unchanged issues do not generate repeated notifications, and that dismissing a suggestion does not modify processing results or workflow decisions.
- VC36.15 – Non- Destructive AI Assistance Verify that conversational interactions, recommendations, and unapproved action proposals do not modify source datasets, processing outputs, analytical results, workflow configurations, or recorded user decisions. Outstanding definitions and implementation decisions Shared identifiers and provenance contract Define mandatory workflowExecutionID, datasetVersionID, operationID, decisionID, resultID, reportSnapshotID; their uniqueness, cardinality, persistence and API representation. Dataset version and lineage semantics Define whether versions are immutable snapshots, how derived datasets reference parent versions, and how re-execution invalidates downstream outputs. Integration eligibility and warnings Define whether and under which explicit approval incomplete acquisitions or validation warnings may proceed; harvesting incompleteness must never silently appear complete. Validation policy and thresholds Specify severity taxonomy, required versus recommended Darwin Core fields, geographic and temporal plausibility rules, unit compatibility, taxonomic resolution and thresholds for blocking. Duplicate detection and conflict resolution Define matching keys, precedence, fuzzy thresholds, cross-source duplicate semantics, conflict resolution options and provenance of merged/retained records. Semantic artefact and taxonomic sources Choose ontology/vocabulary registries, supported versions, licensing, taxonomic backbones, identifier policies and rules for semantic equivalence versus relatedness. Darwin Core target profile Specify core/extension structure (Occurrence, Event, MeasurementOrFact), mandatory fields, identifier scheme, controlled terms and DwC-A packaging. Harmonisation and AI approval rules Define deterministic transformation library, ambiguous value handling, review queue, rollback and user roles/permissions. Analysis catalogue implementation List exact supported methods, required data, aggregation and observational units, versioning of method implementations, missing data handling and sampling-effort limitations. Export and EML metadata Specify required EML profile/version, metadata completeness, validation rules, licences, citation rules, output naming and format constraints. Report lifecycle and consistency Define inclusion rules, snapshot consistency boundaries, skipped stages, status transitions, selection of results, retention and PDF/JSON schemas. AI grounding and governance Define supported actions, evidence retrieval boundaries, model/version logging, confidence/uncertainty presentation, suggestion frequency, permissions and human approval workflow. User access and security Define authentication, authorisation, audit retention, sensitive data handling, API credentials, secure downloads and source repository usage conditions. Performance and non-functional acceptance Specify dataset volumes, timeouts, pagination/retry limits, response times, concurrency, recoverability, availability, logging and operational monitoring. Verification fixtures and acceptance thresholds Create test datasets, reference expected outputs, boundary cases, negative cases and pass/fail thresholds for each verification criterion. Cross-module contract tests Test search-to-harvest links, partial harvest gating, validation-to-harmonisation warnings, transformation lineage, integration-to-analysis applicability, RF34-to-RF35 result preservation and RF36 controlled actions. Terminology and editorial confirmation Confirm source title/content mismatches for

original RF28 and RF29, restored RF28 body from harmonisation reference, and the final disposition of legacy RF16/RF17 versus export requirements.

Items still to be defined

1. Shared identifiers and provenance contract Define mandatory workflowExecutionID, datasetVersionID, operationID, decisionID, resultID, reportSnapshotID; their uniqueness, cardinality, persistence and API representation.
2. Dataset version and lineage semantics Define whether versions are immutable snapshots, how derived datasets reference parent versions, and how re-execution invalidates downstream outputs.
3. Integration eligibility and warnings Define whether and under which explicit approval incomplete acquisitions or validation warnings may proceed; harvesting incompleteness must never silently appear complete.
4. Validation policy and thresholds Specify severity taxonomy, required versus recommended Darwin Core fields, geographic and temporal plausibility rules, unit compatibility, taxonomic resolution and thresholds for blocking.
5. Duplicate detection and conflict resolution Define matching keys, precedence, fuzzy thresholds, cross-source duplicate semantics, conflict resolution options and provenance of merged/retained records.
6. Semantic artefact and taxonomic sources Choose ontology/vocabulary registries, supported versions, licensing, taxonomic backbones, identifier policies and rules for semantic equivalence versus relatedness.
7. Darwin Core target profile Specify core/extension structure (Occurrence, Event, MeasurementOrFact), mandatory fields, identifier scheme, controlled terms and DwC-A packaging.
8. Harmonisation and AI approval rules Define deterministic transformation library, ambiguous value handling, review queue, rollback and user roles/permissions.
9. Analysis catalogue implementation List exact supported methods, required data, aggregation and observational units, versioning of method implementations, missing data handling and sampling-effort limitations.
10. Export and EML metadata Specify required EML profile/version, metadata completeness, validation rules, licences, citation rules, output naming and format constraints.
11. Report lifecycle and consistency Define inclusion rules, snapshot consistency boundaries, skipped stages, status transitions, selection of results, retention and PDF/JSON schemas.
12. AI grounding and governance Define supported actions, evidence retrieval boundaries, model/version logging, confidence/uncertainty presentation, suggestion frequency, permissions and human approval workflow.
13. User access and security Define authentication, authorisation, audit retention, sensitive data handling, API credentials, secure downloads and source repository usage conditions.
14. Performance and non-functional acceptance Specify dataset volumes, timeouts, pagination/retry limits, response times, concurrency, recoverability, availability, logging and operational monitoring.
15. Verification fixtures and acceptance thresholds Create test datasets, reference expected outputs, boundary cases, negative cases and pass/fail thresholds for each verification criterion.
16. Cross-module contract tests Test search-to-harvest links, partial harvest gating, validation-to-harmonisation warnings, transformation lineage, integration-to-analysis applicability, RF34-to-RF35 result preservation and RF36 controlled actions.
17. Terminology and editorial confirmation Confirm source title/content mismatches for original RF28 and RF29, restored RF28 body from harmonisation reference, and the final disposition of legacy RF16/RF17 versus export requirements.